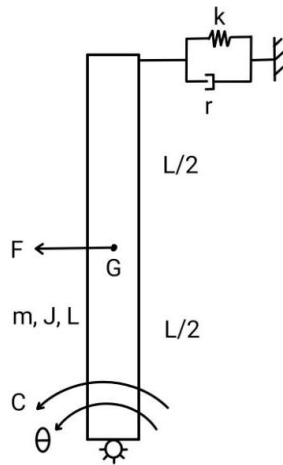


Semester: 2025/2026

Report N°4: DC Motor



Variable	Value
g	9.81 m/s^2
m	80 kg
J	$20 \text{ kg} \cdot \text{m}^2$
L	1 m
r	25 Ns/m
C	[DC Motor Torque]
L_a	0.001 H
R_a	0.1Ω
k_ϕ	0.0001 Nm/A

For this system, the linearized equations of motion and the DC Motor, using the angle ϑ as the independent coordinate, result in:

$$\left(J + m \frac{L^2}{4} \right) \ddot{\theta} + rL^2 \dot{\theta} + \left(kL - \frac{mg}{2} \right) L\theta = k_\phi i_a$$

$$L_a \frac{di_a}{dt} + R_a i_a = V_a - k_\phi \dot{\theta}$$

where the gravitational force on the bar had to be linearized around the equilibrium position of $\vartheta = 0$. In this case k_ϕ is the motor torque constant, V_a is the armature voltage, i_a is the armature current, L_a is the armature inductance, and R_a the armature resistance.

Defining the equivalent mass, damping and stiffness of the system as:

$$m^* = \left(J + m \frac{L^2}{4} \right)$$

$$r^* = (rL^2)$$

$$k^* = \left(kL - \frac{mg}{2} \right) L$$

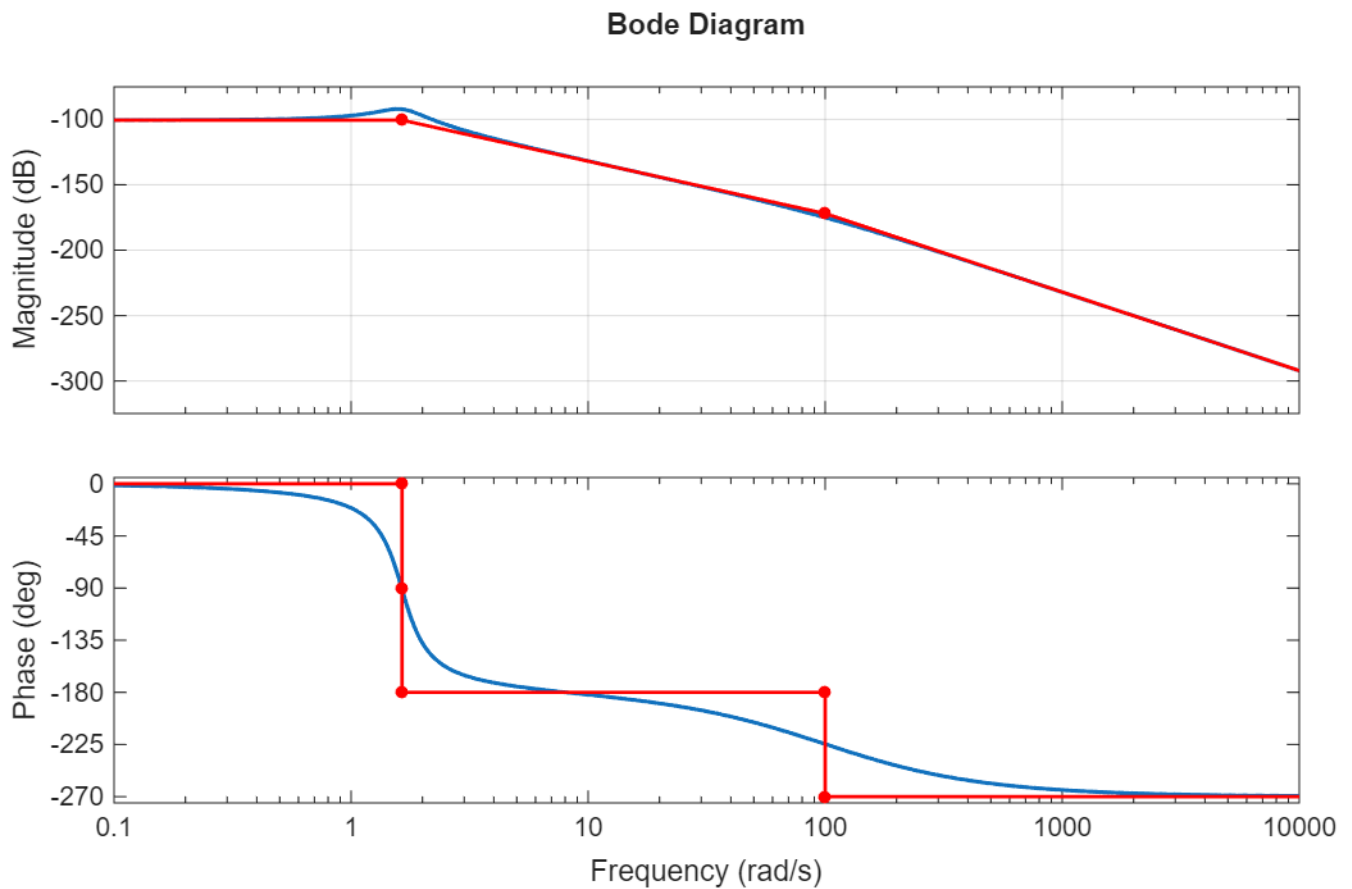
Transforming the equations in the Laplace domain, and operating between them, the transfer function between the armature input voltage and the rotation of the bar can be calculated:

$$G(s) = \frac{\Omega}{V_a} = \frac{k_\phi}{(m^* s^2 + r^* s + k^*)(L_a s + R_a) + k_\phi^2}$$

This transfer function has 3 poles, one stable single pole and 2 complex conjugate stable ones:

$$p_1 = -100 \wedge p_{2,3} = -0.3125 \pm 1.6101j$$

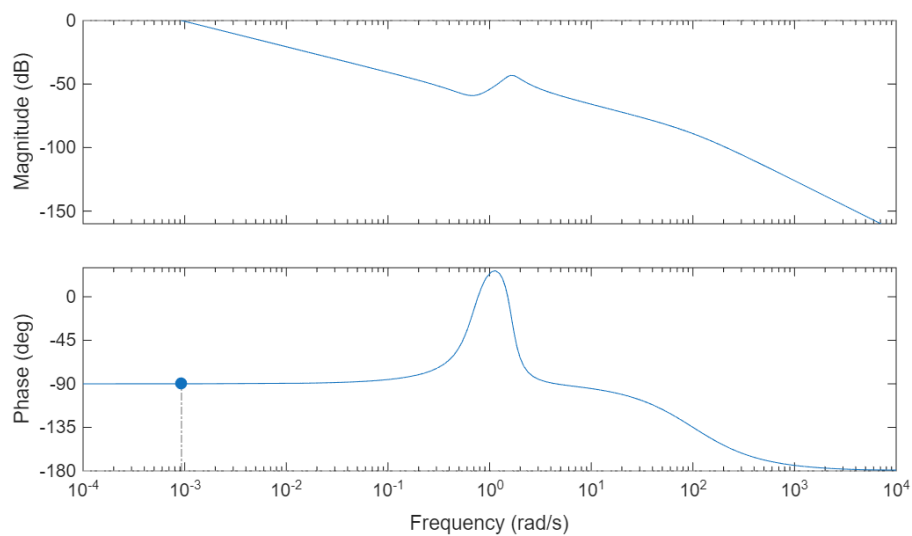
And has a Bode Diagram as such:

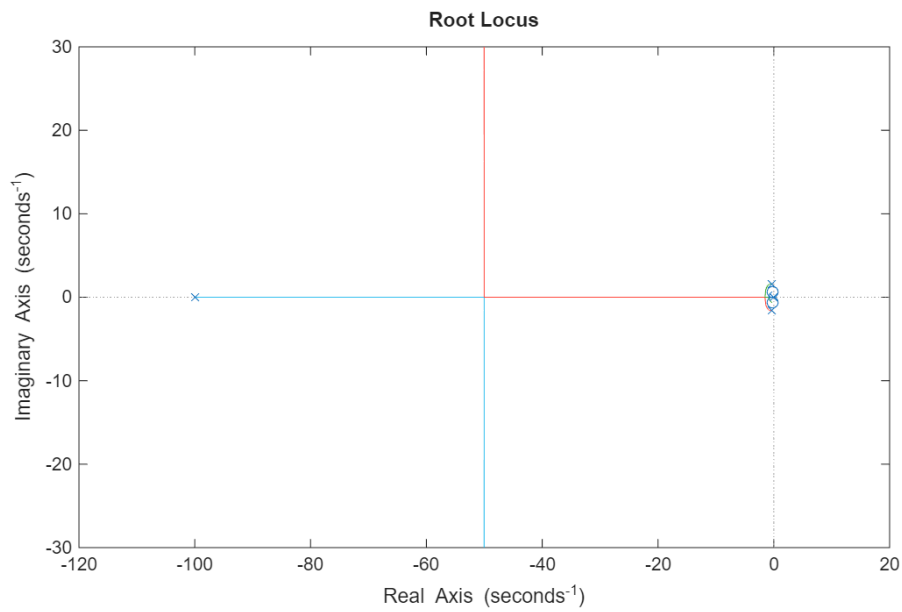


Next, the analysis of different values of T_d and T_i based on their margins, bode and root locus diagrams (everything with a set $k_p = 100V$).

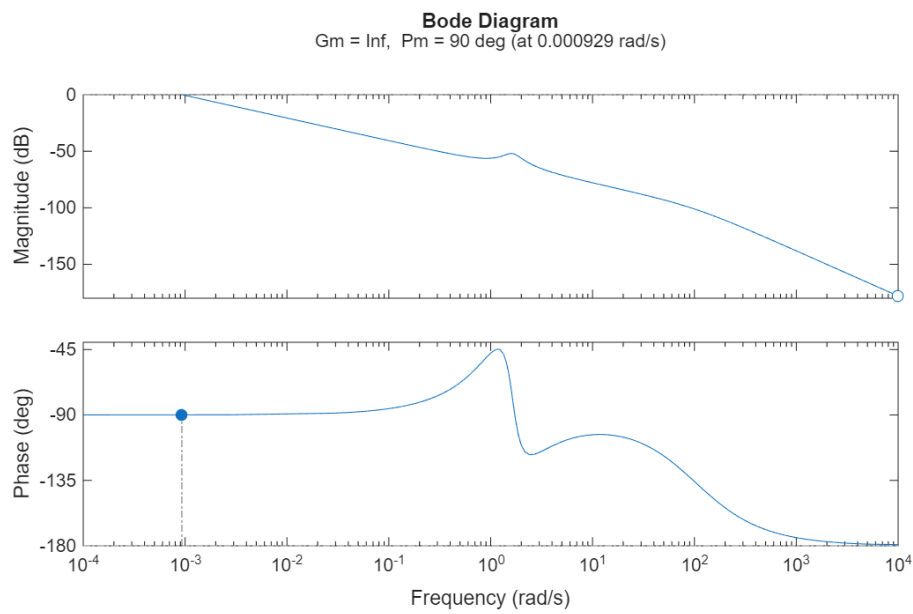
$$T_i = 1s \wedge T_d = 1s$$

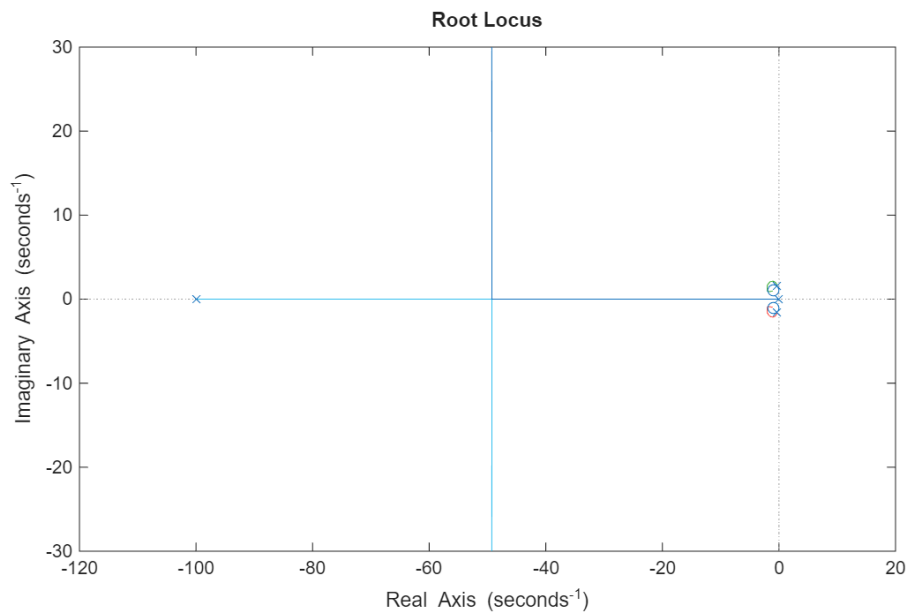
Bode Diagram
Gm = Inf, Pm = 90 deg (at 0.000929 rad/s)



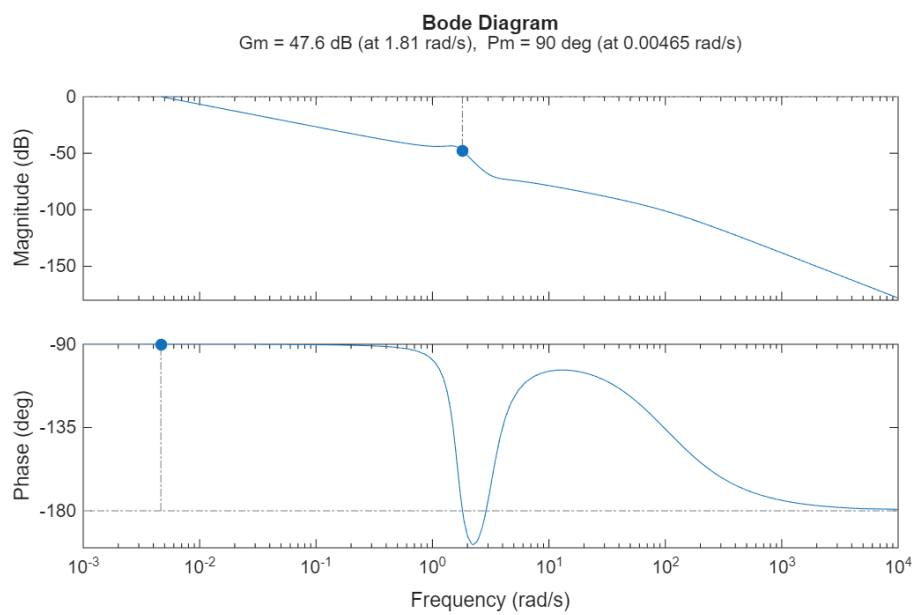


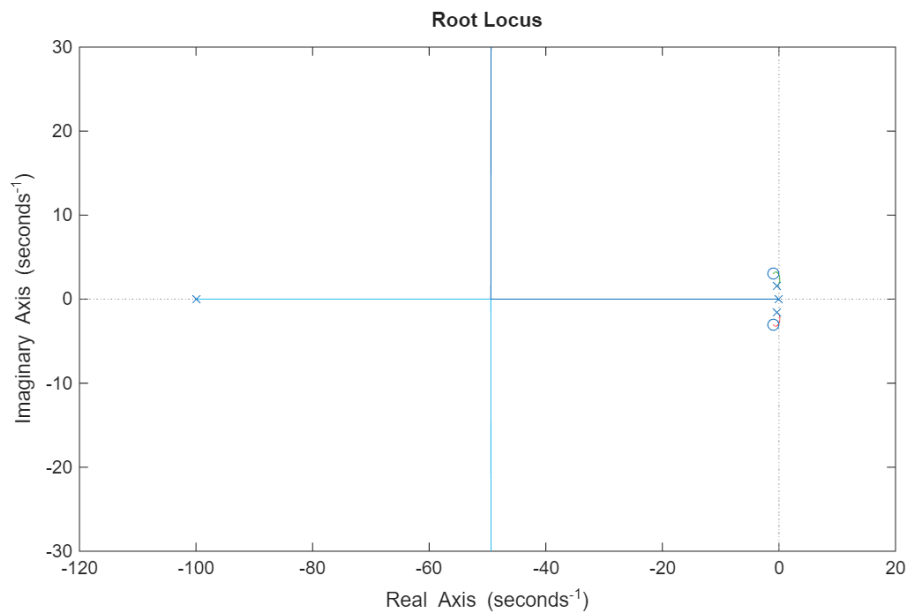
$T_i = 1s \wedge T_d = 0.5s$



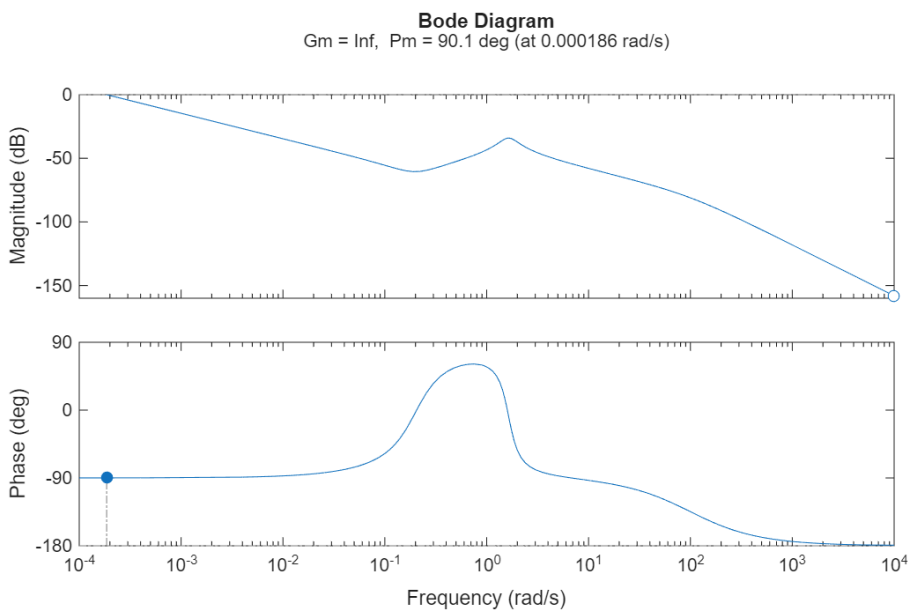


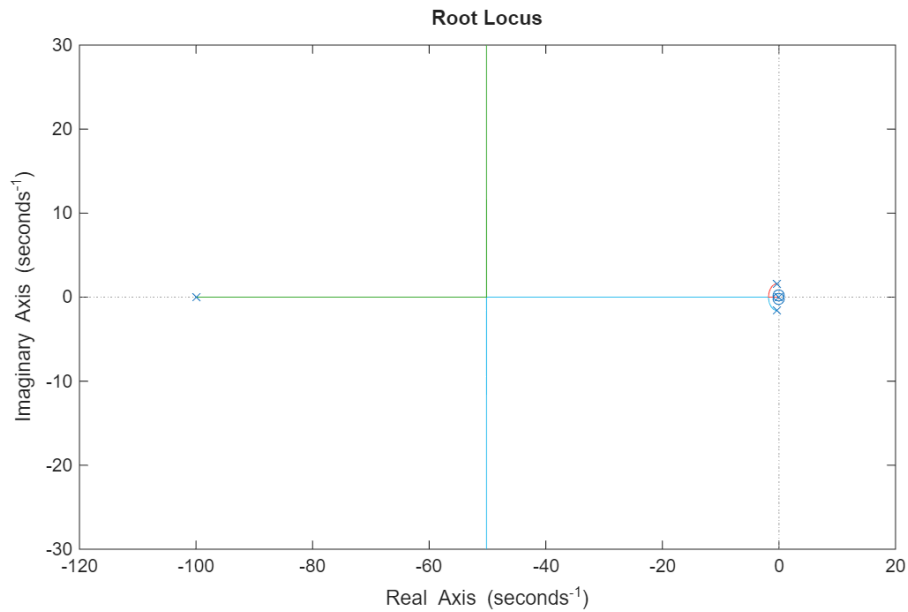
$$T_i = 0.2s \wedge T_d = 0.5s$$





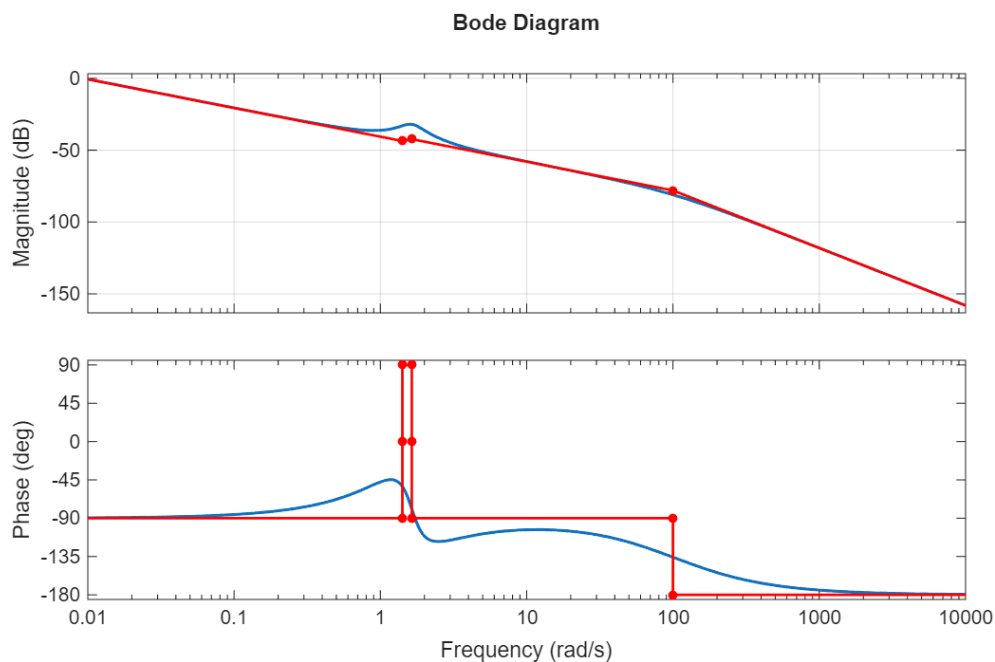
$T_i = 5s \wedge T_d = 5s$



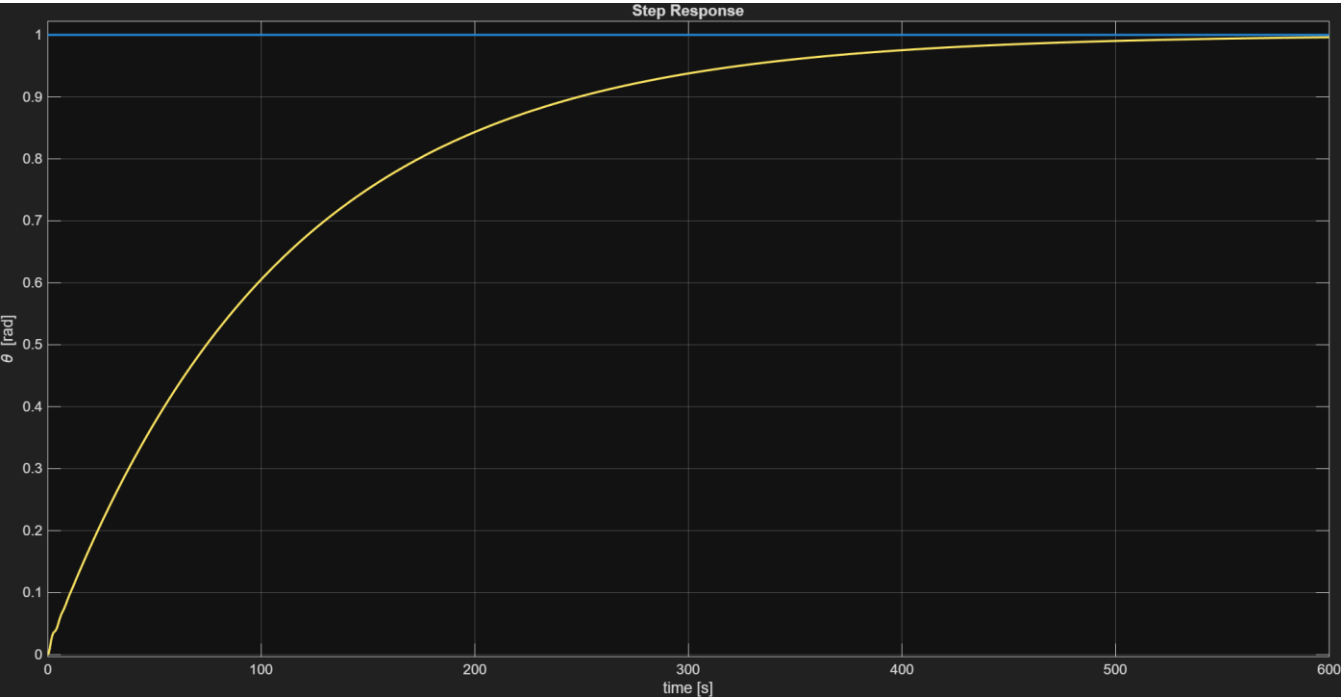


The selection of time constants then is setting them in $T_i = 1s \wedge T_d = 0.5s$. This is because they give a very good phase margin, that isn't sacrificed by increasing the crossover frequency with a larger proportional gain. We're also sure that the system will be stable no matter the proportional gain, because the gain margin is infinite. Moreover, those constants give a faster response than all the other with the benefit of also having infinite gain margin, while staying in a reasonably better phase lag range. With a sufficiently large proportional gain, the response could be fast for low frequencies, the bandwidth high and all of the closed-loop transfer function poles real and negative.

Because of the low torque constant, a high proportional gain is needed to have a good response. The proportional gain will then be set in $k_p = 1000V$. And then, the bode diagrams of the open-loop transfer function becomes.



Lastly, the performance of the feedback control system with these parameters is analysed through its plot and time performance indexes on step response.



With:

Rise Time	238.116 s
Settling Time	423.156 s
Overshoot	0 rad
Steady-State Error	0 rad